Test Plan Document

Web-Based ML Model Evaluator | v1.1 (Corrected)

1. Purpose and Scope

This test plan defines the verification approach for the Web-Based ML Model Evaluator. It maps test cases directly to the FR and NFR identifiers in the companion SRS. The plan covers account access, dataset management, preparation, model training, evaluation, comparison, exports, run history, API behaviour, security, usability, and reliability. Test execution has not been represented as completed in this document. The Actual Result and Test Result columns are intentionally left blank for the team to fill in during execution.

3.1 In Scope

Registration, login, logout, and protected-resource access. CSV upload, validation, preview, statistics, metadata, and dataset replacement. Preprocessing, target selection, train-test split, algorithm selection, and training. Metrics, evaluation visualizations, model comparison, downloads, and run history. API response behaviour, user isolation, reproducibility metadata, browser UI, and failure handling. **Role-based access control: Admin user management, Admin dataset management, Admin log access, and Instructor read-only evaluation access. ★ Added in v1.1.**

3.2 Out of Scope

Testing algorithms or ML libraries independently of their integration with this application. Production-scale penetration testing or formal certification. Formats, algorithms, or export types not implemented in the delivered version.

2. Test Objectives

- Verify each functional requirement through positive, negative, integration, or end-to-end tests.
- Verify important quality requirements, especially authorization, user data isolation, data integrity, and error handling.
- Confirm that invalid data and configuration are rejected safely and consistently.
- Maintain traceability from SRS requirement IDs to test case IDs and outcomes.

4. Test Approach

Test type	Purpose
Unit Testing (UT)	Test individual validation functions, service methods, metrics calculations, and configuration checks.
Integration Testing (IT)	Test interactions among UI/API, backend, database, file storage, and ML components.
System Testing (ST)	Test complete user workflows and cross-cutting requirements from the user's perspective.
Non-functional Testing (NFT)	Test performance, security, compatibility, reliability, and usability where applicable.

5. Test Case ID Convention

Test Case ID format: <Requirement-ID>-<Test-Type>-<Sequence>, for example FR-2.1-UT-1 or NFR-4-ST-1. Requirement-ID is the exact FR or NFR identifier defined in the SRS. Test-Type: UT = Unit Test, IT = Integration Test, ST = System Test, NFT = Non-Functional Test. Sequence is a running number for the selected requirement and test type.

6. Test Environment and Test Data

- Run the application in the project's configured development/test environment with backend, database, storage, and ML dependencies available.
- Use dedicated test accounts (User A and User B) to verify user isolation.
- **Create dedicated role-specific test accounts: one Admin account and one Instructor account for role-based access control testing. ★ Added in v1.1.**
- Prepare valid CSV data, an empty CSV, a malformed CSV, a file with missing values, a dataset with numeric and categorical columns, and a dataset with a usable target column.

- Use small deterministic datasets for metric and reproducibility checks; use a larger within-limit dataset for performance tests.
- Do not use real personal or confidential data in test fixtures.

7. Test Execution Summary

The summary below is a planned test inventory, not a claim that tests have already passed.

Requirement ID	Test cases planned	Coverage area
FR-1.1	2	Register with valid required account fields
FR-1.2	2	Authenticate with valid credentials
FR-1.3	1	Logout terminates the active session
FR-1.4	2	Unauthenticated user cannot access protected pages or APIs
FR-2.1	2	Upload a valid CSV dataset
FR-2.2	2	Reject an empty or malformed CSV
FR-2.3	1	Preview displays correct headers and sample rows
FR-2.4	1	Dataset statistics match a known dataset
FR-2.5	2	Replacing a dataset updates the current user's selected dataset
FR-3.1	2	Apply a supported preprocessing option to valid data
FR-3.2	2	Select a valid target column
FR-3.3	2	Accept a valid train-test split
FR-4.1	2	Display supported algorithms and accept a compatible selection
FR-4.2	2	Complete training for a valid configuration
FR-4.3	3	Calculate applicable metrics for classification
FR-4.4	2	Compare two completed runs
FR-5.4	2	Download a supported model artifact or report
FR-6.1	2	Show a user's prior runs and configuration
FR-6.2	2	API returns a successful structured response for a valid request
FR-6.3	1	Show actionable errors for a failed operation
FR-7.1	3	Admin can view and manage user accounts; non-Admin is denied
FR-7.2 ★	3	Admin can view and manage dataset records; non-Admin is denied
FR-7.3 ★	3	Admin can view activity logs; non-Admin and Instructor are denied
FR-7.4 ★	3	Instructor can view authorized evaluation results in read-only mode
FR-7.5 ★	2	Role-based access control is enforced server-side for all roles
NFR-1	1	Measure response time for common preview/list actions
NFR-2	1	Show progress/status for a long-running training job
NFR-3	1	Verify passwords are not stored in plaintext
NFR-4	1	Enforce authorization on direct backend requests
NFR-5	1	Verify isolation across datasets, runs, and artifacts
NFR-6	1	Preserve completed results after a later operation fails
NFR-7	1	Retain run configuration needed to explain a result
NFR-8	1	Verify the primary workflow is understandable in the desktop UI
NFR-9	1	Check core workflows in supported browsers
NFR-10	1	Verify separation of major application components

Requirement ID	Test cases planned	Coverage area
NFR-11	1	Verify upload/resource limits are enforced
NFR-12	1	Verify run metadata and artifacts map to the correct run
NFR-13	1	Ensure errors do not reveal sensitive internals
NFR-14	1	Deploy from documented setup instructions
TOTAL	61 ★	Planned test cases; execute and record actual outcomes

★ Rows marked with ★ are new or updated in v1.1. Previous total was 51.

8. Detailed Test Cases

For each case, record Actual Result and Test Result after execution. Use Pass, Fail, Blocked, or Not Run as appropriate.

FR-1.1

Test Case ID	Module	Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-1.1-UT-1	3.1 User Access	Register with valid required fields	Registration page is available.	1. Open registration. 2. Enter valid unique credentials. 3. Submit.	Valid unique username/email; valid password	Account is created and success feedback is shown.		
FR-1.1-UT-2	3.1 User Access	Reject registration with missing or invalid fields	Registration page is available.	1. Leave a required field blank or enter invalid data. 2. Submit.	Blank field / invalid email / out-of-bounds password	Validation is shown, no account is created.		

FR-1.2

Test Case ID	Module	Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-1.2-UT-1	3.1 User Access	Authenticate with valid credentials	Registered test account exists.	1. Open login. 2. Enter valid credentials. 3. Submit.	Valid username and password	User is authenticated and redirected to the application.		
FR-1.2-UT-2	3.1 User Access	Reject invalid credentials	Registered test account exists.	1. Enter incorrect password. 2. Submit.	Wrong password	Login is rejected without exposing account details.		

FR-1.3

Test Case ID	Module	Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-1.3-IT-1	3.1 User Access	Logout terminates the active session	User is logged in.	1. Select Logout. 2. Navigate back to a protected page.	Authenticated test account	Session ends and protected content is not accessible.		

FR-1.4

Test Case ID	Module	Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-1.4-ST-1	3.1 User Access	Unauthenticated user cannot access protected pages or APIs	User is logged out.	1. Open a protected URL or call a protected endpoint directly.	No session/token	Access is denied or redirected to login; no protected data is returned.		
FR-1.4-ST-2	3.1 User Access	Authenticated user cannot access another user's resources	User A and User B exist; User B owns a dataset/run.	1. Log in as User A. 2. Request User B's resource identifier.	User B dataset/run ID	The request is denied and no resource details or artifacts are disclosed.		

FR-2.1

Test Case ID	Module	Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-2.1-UT-1	3.2 Dataset Mgmt	Upload a valid CSV dataset	User is authenticated.	1. Open dataset upload. 2. Select valid CSV. 3. Submit.	CSV with headers and data rows	Upload succeeds and the dataset is associated with the authenticated user.		

Test Case ID	Module	Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-2.1-UT-2	3.2 Dataset Mgmt	Reject an unsupported file type	User is authenticated.	1. Select a non-CSV file. 2. Submit.	PDF or executable extension	Upload is rejected with a clear error message.		

FR-2.2

Test Case ID	Module	Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-2.2-UT-1	3.2 Dataset Mgmt	Reject malformed or empty CSV	Upload page is available.	1. Upload an empty or malformed CSV. 2. Submit.	Empty file / malformed data	Validation fails and unacceptable data is not stored.		
FR-2.2-IT-2	3.2 Dataset Mgmt	Validate required tabular structure	User is authenticated.	1. Upload a CSV with inconsistent column counts or no data rows. 2. Review response.	Misformed headers or missing rows	System reports the issue and does not proceed with the dataset.		

FR-2.3

Test Case ID	Module	Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-2.3-ST-1	3.2 Dataset Mgmt	Preview displays correct headers and sample rows	A valid dataset is uploaded.	1. Open dataset preview. 2. Compare displayed values.	CSV with known headers and sample values	Preview shows the correct column names and rows without order or column shifts.		

FR-2.4

Test Case ID	Module	Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-2.4-UT-1	3.2 Dataset Mgmt	Dataset statistics match a known dataset	A valid dataset with known dimensions is uploaded.	1. Open dataset statistics. 2. Compare stats.	10 rows, 3 columns, known missing values	Displayed dimensions and available information match the fixture.		

FR-2.5

Test Case ID	Module	Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-2.5-IT-1	3.2 Dataset Mgmt	Replacing a dataset updates the current user's selected dataset	User has already uploaded Dataset A.	1. Upload Dataset B. 2. Open metadata and review.	Dataset A: 5 rows; Dataset B: 8 rows	Current dataset reflects Dataset B; Dataset A's data is not shown.		
FR-2.5-ST-2	3.2 Dataset Mgmt	Dataset metadata is isolated by owner	User A and User B each have a dataset.	1. Sign in as each user. 2. Open dataset listing/history.	Different dataset names per user	Each user sees only their own datasets and never the other's.		

FR-3.1

Test Case ID	Module	Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-3.1-UT-1	3.3 Data Prep	Apply a supported preprocessing option to valid data	A dataset with missing values or categorical data is uploaded.	1. Select a supported preprocessing option. 2. Apply it.	Dataset with known missing/categorical values	Preprocessing completes and output conforms to the selected option.		

Test Case ID	Module	Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-3.1-UT-2	3.3 Data Prep	Reject incompatible preprocessing configuration	A dataset is uploaded.	1. Select an incompatible option. 2. Submit.	Invalid option/parameter combination	Invalid configuration is rejected with a useful message.		

FR-3.2

Test Case ID	Module	Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-3.2-UT-1	3.3 Data Prep	Select a valid target column	Dataset has multiple columns.	1. Open target selector. 2. Select an existing column name.	Target column: outcome	Selection is saved and used to configure the model.		
FR-3.2-UT-2	3.3 Data Prep	Reject a missing or invalid target selection	Dataset is uploaded.	1. Leave target unset or provide a nonexistent column name. 2. Confirm.	No column / nonexistent column	Training cannot start and the user is told to select a valid target.		

FR-3.3

Test Case ID	Module	Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-3.3-UT-1	3.3 Data Prep	Accept a valid train-test split	Dataset and target are selected.	1. Choose supported proportions. 2. Save or confirm.	80% train / 20% test	Configuration is accepted and partitions are created with proportional row counts.		
FR-3.3-UT-2	3.3 Data Prep	Reject invalid train-test split	Dataset and target are selected.	1. Enter invalid proportions or values. 2. Submit.	Negative, >100%, or empty value	Configuration is rejected with a clear validation message.		

FR-4.1

Test Case ID	Module	Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-4.1-UT-1	3.4 Model Config	Display supported algorithms and accept a compatible selection	Dataset and task configuration are available.	1. Open algorithm selection. 2. Select a compatible algorithm.	One supported classification or regression algorithm	Selected algorithm is stored in the run configuration.		
FR-4.1-IT-2	3.4 Model Config	Prevent incompatible algorithm/task configuration	Dataset/task type is known.	1. Attempt to select or configure an incompatible algorithm.	Algorithm mismatched with task type	System prevents incompatible execution or returns a clear error.		

FR-4.2

Test Case ID	Module	Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-4.2-IT-1	3.4 Model Training	Complete training for a valid configuration	Valid dataset, target, split, and algorithm are configured.	1. Start training. 2. Wait for completion.	Small deterministic dataset; supported algorithm	Training completes and results are stored for the user.		
FR-4.2-IT-2	3.4 Model Training	Handle training failure without corrupting prior runs	A previous completed run exists.	1. Start a run with invalid or deliberately failing configuration. 2. Review status and prior run history.	Invalid configuration or controlled failure	Failure is reported; previous completed run history remains intact.		

FR-4.3

Test Case ID	Module	Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-4.3-UT-1	3.5 Evaluation	Calculate applicable classification metrics	A classification model has predictions and labels.	1. Evaluate the run. 2. Compare metrics with known labels.	Known labels and expected classification metrics	Applicable classification metrics match the expected calculation.		
FR-4.3-UT-2	3.5 Evaluation	Calculate applicable regression metrics	A regression model has predictions and labels.	1. Evaluate the run. 2. Compare metrics with known numeric labels.	Known labels and expected regression metrics	Applicable regression metrics match expected values within tolerance.		
FR-4.3-ST-3	3.5 Evaluation	Display evaluation visualizations where applicable	A completed run exists.	1. Open results. 2. Inspect charts/plots.	Classification run or regression run	Relevant visualizations are shown and correspond to the selected run.		

FR-4.4

Test Case ID	Module	Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-4.4-IT-1	3.5 Comparison	Compare two completed runs	At least two completed runs with compatible task/metric types exist.	1. Select two runs. 2. Compare.	Two compatible runs	Comparison shows run identities/configurations and compares metrics side by side.		
FR-4.4-UT-2	3.5 Comparison	Prevent misleading comparison of incompatible metrics	Runs of incompatible task types exist.	1. Select one classification and one regression run for comparison.	One classification, one regression	System separates incompatible metrics or explains why direct comparison is unavailable.		

FR-5.4

Test Case ID	Module	Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-5.4-IT-1	3.5 Export	Download a supported model artifact or report	A completed run with a downloadable artifact/report exists.	1. Select download. 2. Open the downloaded file.	Completed run ID	Download succeeds; file is readable and corresponds to the run.		
FR-5.4-UT-2	3.5 Export	Handle unavailable artifact	A run has no artifact of the requested type.	1. Request the download.	Missing artifact type	A clear error is shown and no file is returned.		

FR-6.1

Test Case ID	Module	Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-6.1-IT-1	3.6 Run History	Show a user's prior runs and configuration	A user has at least one completed run.	1. Open run history. 2. Select a prior run.	Run with known algorithm, split, and status	History shows the run's algorithm, split, and status/results correctly.		
FR-6.1-ST-2	3.6 Run History	Keep a user's prior run history private	Two users have separate completed runs.	1. Sign in as User A. 2. Inspect history and request User B's run.	User B run ID	User A cannot see or retrieve User B's run.		

FR-6.2

Test Case ID	Module	Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-6.2-IT-1	3.6 API	API returns a successful structured response for a valid request	Backend service is running.	1. Call a documented endpoint with valid input. 2. Inspect response.	Valid request payload	Expected success status and structured response are returned.		
FR-6.2-IT-2	3.6 API	API returns an error for an invalid or incomplete request	Backend service is running.	1. Call a documented endpoint with invalid or incomplete input.	Malformed/missing field	A suitable error status and structured error response are returned.		

FR-6.3

Test Case ID	Module	Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-6.3-ST-1	3.6 Error Handling	Show actionable errors for a failed operation	Application is available.	1. Trigger an invalid upload or run. 2. Review UI feedback.	Malformed CSV or invalid config	Message explains the problem without exposing internal details or stack traces.		

NFR-1

Test Case ID	Module	Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
NFR-1-NFT-1	4. Non-Functional	Measure response time for common preview/list actions	Test environment and representative data are ready.	1. Run the same preview/list request repeatedly under normal load. 2. Record response times.	Documented dataset size	Response-time measurements are recorded and compared to the project target.		

NFR-2

Test Case ID	Module	Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
NFR-2-NFT-1	4. Non-Functional	Show progress/status for a long-running training job	A configuration is available that takes longer than a normal request.	1. Start training. 2. Observe UI while it runs.	Long-running but valid configuration	The user receives running/progress/status feedback while the job runs.		

NFR-3

Test Case ID	Module	Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
NFR-3-ST-1	4. Non-Functional	Verify passwords are not stored in plaintext	A test account can be accessed and the datastore is inspectable.	1. Register a test account. 2. Inspect the stored representation in the test database.	Test account password	Stored value is a secure hash, not the submitted plaintext.		

NFR-4

Test Case ID	Module	Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
NFR-4-ST-1	4. Non-Functional	Enforce authorization on direct backend requests regardless of UI	User A and User B each have resources.	1. Authenticate as User B. 2. Call the backend directly for User A's resource.	User B identifier	Backend denies access; resource is not returned.		

NFR-5

Test Case ID	Module	Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
NFR-5-ST-1	4. Non-Functional	Verify isolation across datasets, runs, and artifacts	Two users have completed uploads and runs.	1. Sign in as each user. 2. Check dataset and run lists and direct source IDs.	Distinct user-owned resources	No cross-user resources are visible or accessible.		

NFR-6

Test Case ID	Module	Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
NFR-6-IT-1	4. Non-Functional	Preserve completed results after a later operation fails	A completed run exists.	1. Record prior result details. 2. Trigger a separate failing operation. 3. Reopen prior run.	Known completed run	Prior result and metadata remain unchanged and accessible.		

NFR-7

Test Case ID	Module	Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
NFR-7-IT-1	4. Non-Functional	Retain run configuration needed to explain a result	A model run completes.	1. Open run details/history. 2. Verify algorithm, target, split, and preprocessing settings.	Known run configuration	Stored details match the settings used for the run.		

NFR-8

Test Case ID	Module	Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
NFR-8-ST-1	4. Non-Functional	Verify the primary workflow is understandable in the desktop UI	Application is deployed.	1. Ask a tester to upload data, configure, train, and view results using UI labels only. 2. Record observations.	Desktop browser	Workflow steps and validation feedback are understandable.		

NFR-9

Test Case ID	Module	Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
NFR-9-NFT-1	4. Non-Functional	Check core workflows in supported browsers	Browser matrix is defined for the release.	1. Execute login, upload, train, and results workflow in each browser in the matrix.	Current agreed versions of Firefox, Edge, Chrome	Core layout and interactions work in each browser; discrepancies are recorded.		

NFR-10

Test Case ID	Module	Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
NFR-10-IT-1	4. Non-Functional	Verify separation of major application components	Source repository and test environment are available.	1. Review module boundaries and imports. 2. Run relevant tests.	Frontend, API/service, persistence, ML components	Responsibilities are separated as documented.		

NFR-11

Test Case ID	Module	Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
NFR-11-NFT-1	4. Non-Functional	Verify upload/resource limits are enforced	Configured limits are known.	1. Upload a file below the limit. 2. Upload a file above the limit.	Below-limit and above-limit file fixtures	Supported input is accepted; over-limit input is rejected with clear feedback.		

NFR-12

Test Case ID	Module	Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
NFR-12-IT-1	4. Non-Functional	Verify run metadata and artifacts map to the correct run	Two runs exist with distinct settings and results.	1. Open each run. 2. Inspect metrics, configurations, and artifact references.	Run A and Run B with different algorithms or targets	Each run displays only its own configuration, metrics, and artifact references.		

NFR-13

Test Case ID	Module	Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
NFR-13-ST-1	4. Non-Functional	Ensure errors do not reveal sensitive internals	A controlled backend error can be triggered in the test environment.	1. Trigger the error. 2. Inspect UI and API response.	Invalid request or controlled error	User-facing response does not contain secrets, passwords, stack traces, or internal configuration.		

NFR-14

Test Case ID	Module	Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
NFR-14-ST-1	4. Non-Functional	Deploy from documented setup instructions	A clean test environment is available.	1. Follow setup instructions. 2. Install declared dependencies and start the application.	Clean environment and declared dependencies	Application starts using documented steps, or missing/incorrect instructions are recorded as defects.		

9. Entry and Exit Criteria

9.1 Entry Criteria

- A testable build is deployed and its required services are running.
- Test accounts, fixture datasets, and required ML dependencies are available.
- **Role-specific test accounts (Admin and Instructor) have been created in the test environment. ★ Added in v1.1.**
- The team has identified the supported algorithms, formats, and browser matrix for the build.

9.2 Exit Criteria

- All planned cases have an execution status recorded.
- Critical access-control, user-isolation, data-loss, and training-workflow defects are resolved or formally documented.
- **All Admin and Instructor role-based access control tests have been executed and outcomes recorded. ★ Added in v1.1.**
- Failed and blocked tests include notes and reproducible steps.
- Known deviations from the SRS and unsupported behaviour are documented for the evaluator.

10. Defect Reporting

Each defect should include a concise title, affected requirement/test case ID, environment, preconditions, reproduction steps, expected result, actual result, severity, evidence (such as screenshot/log with secrets removed), and assignee. Re-test the failed case after a fix and record the new outcome.

11. Requirement Traceability Notes

Every detailed test case uses the requirement ID from the SRS as its prefix. The inventory includes functional and non-functional checks. Some NFRs (for example maintainability and usability) require review or observation in addition to automated tests. The exact numeric performance target, maximum upload size, supported algorithm list, and supported browser versions must be confirmed by the project team and recorded before final acceptance. **FR-7.5 (role-based access control) must be implemented before the FR-7.1, FR-7.2, FR-7.3, and FR-7.4 test cases can be executed, as they depend on a working role field in the user account model. ★ Added in v1.1.**

12. Security Test Cases and Cross-Document Traceability

These cases supplement the functional and non-functional tests above. SEC IDs refer to the companion Security Requirements Document. They are planned tests, not completed results; fill Actual Result and Test Result only after execution. Security test ID convention: <SEC requirement ID>-<test type>-<sequence>. For review-based checks, REV denotes a security configuration/code review.

SEC Req	Test Case ID	Linked SRS ID(s)	Description	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Test Result
SEC-AUTH-1	SEC-AUTH-1-ST-1	FR-1.4	Anonymous access to protected data is denied.	User is logged out.	1. Request a protected dataset, run history, or download endpoint. 2. Inspect response.	No session	Request is denied; no protected content is returned.		
SEC-AUTH-2	SEC-AUTH-2-ST-1	NFR-3	Passwords are stored as salted hashes, not plaintext.	Test registration is enabled; authorized test database is accessible.	1. Register a test user. 2. Inspect the stored password representation.	Known test password	Stored value is a salted password hash and does not equal the plaintext.		
SEC-AUTH-3	SEC-AUTH-3-ST-1	FR-1.2; NFR-13	Login errors do not reveal account existence.	Login endpoint is available.	1. Attempt login with unknown username. 2. Attempt login with wrong password. 3. Compare messages.	Unknown user; wrong password	Messages do not disclose whether an account exists.		
SEC-AUTH-4	SEC-AUTH-4-NFT-1	NFR-3; NFR-11	Repeated failed requests trigger rate limiting or throttling.	Test environment and rate-limit policy are available.	1. Submit repeated failed login requests. 2. Observe server response.	Repeated invalid credentials	Configured throttling /lockout activates according to the documented policy.		
SEC-SES-1	SEC-SES-1-ST-1	FR-1.2; NFR-3	Session handling avoids unnecessary credential exposure.	User can log in; test environment is available.	1. Log in. 2. Inspect URLs, logs, and session tooling.	Approved test account	Credentials/tokens are not exposed in URLs or logs.		
SEC-SES-2	SEC-SES-2-ST-1	FR-1.3; NFR-3	Expired session requires re-authentication.	Session expiry policy is configured or documented.	1. Log in. 2. Wait for or simulate expiry. 3. Request a protected action.	Expired session	Protected action is denied until the user re-authenticates.		
SEC-SES-3	SEC-SES-3-ST-1	FR-1.3	Logout invalidates the session.	User is logged in.	1. Log out. 2. Replay the previous session/token against a protected endpoint.	Old session/token	Previously valid session/token is denied access.		
SEC-AZ-1	SEC-AZ-1-ST-1	FR-1.4; NFR-4	Backend enforces authorization independently of UI.	A protected endpoint exists.	1. Call endpoint directly without authorization. 2. Repeat with unauthorized account.	No session / unauthorized session	Backend denies the request; no protected data is returned.		
SEC-AZ-2	SEC-AZ-2-ST-1	FR-2.5; FR-6.1; NFR-5	User cannot access another user's datasets or runs.	User A and User B each own separate resources.	1. Sign in as User A. 2. Request User B's dataset/run by identifier. 3. Try to download User B's artifact.	User B resource ID	All unauthorized access attempts are denied without leaking resource details.		

SEC Req	Test Case ID	Linked SRS ID(s)	Description	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Test Result
SEC-DAT-1	SEC-DAT-1-ST-1	NFR-5	Check cross-user exposure in listings, history, and direct IDs.	Two users have resources.	1. Compare each user's dataset list and run history. 2. Try direct resource IDs from the other user.	Distinct user resources	Only the owner's resources are visible and accessible.		
SEC-DAT-2	SEC-DAT-2-IT-1	FR-2.1; FR-2.2; NFR-11	Reject malformed, unsupported, and over-limit uploads.	Upload endpoint and documented limits are available.	1. Upload valid CSV. 2. Upload malformed CSV. 3. Upload unsupported format. 4. Upload over-limit file.	Valid; malformed; non-CSV; oversized file	Valid input is accepted; malformed, non-CSV, and oversized input is rejected with clear feedback.		
SEC-DAT-3	SEC-DAT-3-IT-1	FR-4.3; FR-6.1; NFR-6; NFR-12	Failed run does not corrupt a completed run.	A completed run exists.	1. Record prior run details. 2. Trigger a separate failing run. 3. Reopen prior run.	Completed and failing configurations	Prior run details and artifacts remain unchanged.		
SEC-DAT-4	SEC-DAT-4-IT-1	FR-5.4; NFR-12	Downloaded artifact belongs to the selected run.	Two runs have distinct IDs/configurations and available artifacts.	1. Select Run A and download. 2. Select Run B and download. 3. Compare metadata/content.	Run A and Run B	Each download corresponds to the authorized run and content.		
SEC-INP-1	SEC-INP-1-UT-1	FR-2.2; FR-3.1–3.3; FR-4.1; FR-5.4	Backend validates all user-controlled configuration identifiers.	Relevant API endpoints are available.	1. Submit invalid dataset ID, target column, split, algorithm, preprocessing option, and export values directly to the API.	Invalid configuration values	Invalid requests are rejected; no processing or storage occurs.		
SEC-INP-2	SEC-INP-2-REV-1	NFR-10; NFR-13	Review database access for injection resistance.	Source code and test environment are available.	1. Review database access patterns. 2. Submit text containing SQL-like characters to controlled inputs.	Strings containing SQL syntax	Queries treat user input as data; no injection occurs.		
SEC-INP-3	SEC-INP-3-ST-1	NFR-13	User text and errors do not expose internal details.	Text fields and error responses are controllable.	1. Submit script-like text where accepted. 2. Trigger controlled errors. 3. Inspect UI and API response.	Script-like input; invalid requests	Text is safely rendered; errors do not expose stack traces, secrets, or internal paths.		
SEC-OPS-1	SEC-OPS-1-REV-1	NFR-3; NFR-13; NFR-14	No real credentials or secrets are committed to the repository or exposed in builds.	Repository and build artifacts are available.	1. Search tracked files and built frontend for secrets. 2. Verify test-only sentinel credentials are not in the main branch.	Known test-only sentinel values	No real credentials or secrets are present.		

SEC Req	Test Case ID	Linked SRS ID(s)	Description	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Test Result
SEC-OPS-2	SEC-OPS-2-REV-1	NFR-10; NFR-14	Dependencies are declared and a review process is available.	Dependency files are available.	1. Verify dependencies are declared. 2. Inspect requirements/package files. 3. Review the dependency vulnerability-check process if configured.	Requirements documentation	Dependencies are declared and the team can review the vulnerability-scanning process.		
SEC-OPS-3	SEC-OPS-3-REV-1	NFR-6; NFR-14	Recovery steps are documented for persistent data.	The application stores persistent data or artifacts.	1. Review backup and restore instructions. 2. If configured, restore in a safe environment.	Backup/restore documentation	Recovery steps are documented; any unconfigured capability is recorded as a gap.		
SEC-OPS-4	SEC-OPS-4-IT-1	FR-6.3; NFR-13	Security events are logged without sensitive information.	Logging is enabled in the test environment.	1. Trigger failed login and denied access. 2. Inspect the log entries.	Test account; denied resource access	Relevant events are logged without passwords, tokens, or dataset contents.		
SEC-OPS-5	SEC-OPS-5-NFT-1	NFR-3; NFR-14	Deployed credential/data traffic uses HTTPS/TLS.	A deployed test environment is available.	1. Open the application over HTTPS. 2. Inspect connection/security configuration.	Deployed URL	Credentials and uploaded data use HTTPS/TLS; any local-development HTTP exception is documented.		

12.1 Three-Document Traceability Rules

SRS FR/NFR IDs remain the primary functional and quality requirement identifiers. SEC IDs add security-specific detail. Test cases link to both where applicable. If a SEC requirement is marked proposed in the Security Requirements Document, the team should approve it before treating a failed test as a release-blocking defect.

12.2 Security Test Status

No security test is marked Pass by default. Execute the test, record evidence where appropriate, and fill in Actual Result and Test Result. Code-review checks should record reviewer, date, files/areas reviewed, and findings.

13. Additional Tests for Administrative and Instructor Use Cases

These cases align the test plan with UC-09 and UC-12 to UC-14 in the Use Case Flow document, and with FR-7.1 through FR-7.5 in the SRS. They are planned tests; record actual outcomes only after execution. **All tests in this section require FR-7.5 (role-based access control) to be implemented before they can run.**

Test Case ID	Requirement Trace	Description	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Test Result
FR-7.1-ST-1	FR-7.1; SEC-AZ-3	Admin can view the registered users list.	Admin account exists and is authenticated.	1. Open the user-management area. 2. Review the user list.	Several test accounts	The Admin can view the registered user list populated by real accounts.		
FR-7.1-ST-2	FR-7.1; SEC-AZ-3	Admin can perform a supported account management action.	Admin account exists and a non-admin test account exists.	1. Select a test account. 2. Perform a supported management action (e.g., view or deactivate). 3. Record the action confirmation.	Dedicated test account	The action is validated, confirmed, and reflected in the account list.		
FR-7.1-ST-3	FR-7.1; SEC-AZ-3	Non-Admin cannot access or call the user-management area.	A Student and an Instructor test account exist.	1. Log in as a regular user. 2. Open the user-management URL or call the management API directly.	Regular-user session	The request is denied for both the UI and the direct API call.		
FR-7.2-ST-1	FR-7.2; SEC-AZ-4	Admin can view stored dataset records and metadata.	Admin account exists and datasets have been uploaded by test users.	1. Open dataset management area. 2. Inspect a stored dataset record.	Test datasets with known metadata	The Admin view shows accurate dataset owner, row count, column count, and the supported metadata.		
FR-7.2-ST-2	FR-7.2; SEC-AZ-4	Dataset management preserves ownership associations.	Admin account exists and datasets have been uploaded.	1. Inspect a dataset record. 2. Perform a supported management operation.	Dataset associated with a test user	The operation follows permissions and does not alter the dataset's user_id or disassociate it from its model runs.		
FR-7.2-ST-3	FR-7.2; SEC-AZ-4	Non-Admin cannot access administrative dataset-management functions.	A Student and an Instructor test account exist.	1. Log in as a regular user or Instructor. 2. Call the dataset-management API endpoint directly.	Regular-user or Instructor session	The request is denied; no administrative view of other users' dataset records is returned.		
FR-7.3-ST-1 ★	FR-7.3; SEC-AUD-1	Admin can view application activity logs.	Admin account exists and system activity has been recorded.	1. Open the system-log interface as Admin. 2. Review available log entries.	Log entries from uploads and training runs	Admin can view relevant log events; log entries do not expose passwords, session tokens, or raw dataset contents.		
FR-7.3-ST-2 ★	FR-7.3; SEC-AUD-1	Activity logs are read-only and do not expose sensitive data.	Admin account exists and log entries are available.	1. Review log entries as Admin. 2. Confirm no write or delete operations are exposed. 3. Confirm no sensitive data appears in log content.	Log entries containing security events	Logs display only non-sensitive event data; no edit or delete controls are available through the log interface.		

Test Case ID	Requirement Trace	Description	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Test Result
FR-7.3-ST-3 ★	FR-7.3; SEC-AZ-6	Non-Admin and Instructor cannot access the system-log interface or its endpoints.	A Student and an Instructor test account exist.	1. Log in as a Student. 2. Call the log API endpoint directly. 3. Log in as an Instructor. 4. Call the log API endpoint directly.	Student session; Instructor session	Both requests are denied with an authorization error; no log data is returned.		
FR-7.4-ST-1 ★	FR-7.4; SEC-AZ-5	Instructor can view evaluation results that have been made available for review.	Instructor account exists and evaluation results have been explicitly made available for the Instructor to view.	1. Log in as Instructor. 2. Open the evaluation results interface. 3. Review an available result.	Instructor session; available evaluation result	Instructor can see the result details in read-only mode with no modification controls available.		
FR-7.4-ST-2 ★	FR-7.4; SEC-AZ-5	Instructor cannot modify, delete, or create any resource.	Instructor account is authenticated.	1. Attempt to upload a dataset. 2. Attempt to start training. 3. Attempt to modify or delete an evaluation result.	Instructor session	All write, delete, and upload operations are denied for the Instructor role.		
FR-7.4-ST-3 ★	FR-7.4; SEC-AZ-5	Instructor cannot access evaluation results not made available for review.	Instructor account exists and a result not shared with the Instructor exists.	1. Log in as Instructor. 2. Attempt to access the non-shared evaluation result by direct ID.	Instructor session; non-shared result ID	Access is denied; no result details are returned.		
FR-7.5-ST-1 ★	FR-7.5; NFR-4	Role-based access control is enforced server-side for all three roles.	Student, Instructor, and Admin accounts exist.	1. For each role, call endpoints that belong to a higher-privileged role directly via the API.	Student trying Admin endpoint; Instructor trying Admin endpoint	All cross-role API calls are denied by the backend regardless of UI state.		
FR-7.5-IT-1 ★	FR-7.5; FR-1.1	Newly registered users are assigned the Student role by default.	Registration is functional.	1. Register a new user. 2. Inspect the stored role in the database or via the session endpoint.	New user with no explicit role set	The new user's role is Student; they cannot access Admin or Instructor-only endpoints.		

★ Test cases marked with ★ are new in v1.1. They were missing from the original Test Plan.